ASEAN
GEOAI FUSION 2026

Module 4:
Foundational Knowledge: Foundation in GEOAI (FD1004)

AGAIF 2026 Boot Camp 1
Level: Foundation of GeoAI

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From Connectivity Gaps to Evidence-Based Decisions

The learning journey across Sessions 4–6

SESSION 4

UNDERSERVED COMMUNITIES' IDENTIFICATION

UNDERSTAND & IDENTIFY

Who may be underserved — and why?

Combine telecommunications conditions with population, community and spatial context.

Telecom Network · Population · Communities · Public Services

Potential underserved communities

SESSION 5

CASE STUDY – ASEAN TELECOMMUNICATIONS CHALLENGE

COMPARE & REASON

When connectivity needs compete, what evidence should guide priority?

Examine ASEAN telecommunications challenges, evidence layers and competing planning considerations.

Impact · Urgency · Feasibility · Public Value

Transparent priority reasoning

SESSION 6

HANDS-ON

APPLY & RECOMMEND

How can spatial evidence be transformed into a defensible recommendation?

Apply the guided spatial screening workflow, interpret the evidence and extend it into priority and intervention recommendations.

Jupyter Workflow · Spatial Screening · Candidate Areas · Recommendation

Evidence-based decision support

Why This Module Matters

Connectivity inequality is not only a network problem. It is a spatial problem involving people, place, public services and planning priorities.

GeoAI helps connect these evidence layers so that connectivity decisions can be explained and defended.

IDENTIFY → **COMPARE** → **APPLY** → **RECOMMEND**

Learning progression: Understand the problem → Reason with evidence → Apply the workflow → Support a decision

Session 6:
Hands-On

Hands-On Workflow

Checkpoint journey before and during the guided Jupyter notebook

0 – Checkpoint 0 – Prepare the notebook

Set up the geospatial analysis environment

1 – Checkpoint 1 – Define the Sarawak study case

Set the study area, data sources and evidence filters

2 – Checkpoint 2 – Mobile performance observations

Where was lower mobile performance observed?

3 – Checkpoint 3 – Community locations

Who is there? Add mapped communities

4 – Checkpoint 4 – Population exposure

How many people may be spatially associated?

5 – Checkpoint 5 – Public-facility context

Add schools and health facilities

6 – Checkpoint 6 – Screening candidates

Identify potential communities for further connectivity investigation

7 – Checkpoint 7 – Mini intelligence dashboard

Summarise key evidence and interpret the findings

Outcome:

Evidence-based spatial screening insight

GeoAI does not make the decision for us.

It helps organise spatial evidence so that the decision can be explained, compared and defended.

TELECOMMUNICATIONS EVIDENCE

What is the observed network condition?



Mobile performance observations highlight locations of lower performance.

GEOSPATIAL CONTEXT: COMMUNITIES

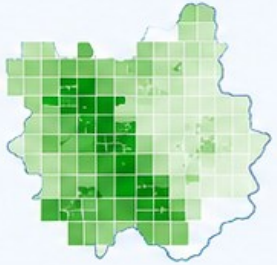
Who is there?



Mapped communities show who may be spatially associated with the network condition.

POPULATION EXPOSURE

How many people may be spatially associated?



Population estimates help indicate the potential number of people affected.

PUBLIC SERVICE CONTEXT

What community-service context exists?



Schools, health facilities and other services provide additional community context.

SCREENING CANDIDATES

Where should further connectivity investigation begin?



Proximity analysis highlighted potential screening candidates for further investigation.

KEY INSIGHT

Screening identifies where further investigation should begin.

Better decisions come from combining network evidence, geospatial context and transparent planning judgement.

GeoAI brings the evidence together. | **People make the informed decision.**

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026